COSC 4368 Assignment 2 Team Contributions

Team Member Name	Primary Responsibilities/Contributions	Percent Contributed
Tyler Landgrebe (2247934)	Set up the local Python/PyTorch environment and documented dependencies. Built and checked the training and validation data pipeline, including image loading, class-label mapping, transforms, and DataLoaders. Served as submission lead. Ran accuracy experiments on a separate GitHub branch, comparing 45-epoch training and horizontal-flip test-time augmentation with the team's 60-epoch model in attempts to increase model accuracy.	20%
Dexter Green (2134291)	Implemented PyTorch training/validation loops, reproducibility controls, learning rate scheduling, and checkpoint saving. Conducted hyperparameter tuning and augmentation experiments, tracked model performance, and improved validation accuracy through fine-tuning. Built 60 epoch training pipeline using AdamW, Cutmix, label smoothing, and Exponential Moving Average in attempts to increase models validation accuracy.	30%
Chase Beeler (2418063)	Designed and iterated on the custom CNN architecture, comparing network depth, channel widths, pooling, dropout, and normalization. Consolidated the team's training code into one pipeline. Extended the team's 60-epoch AdamW, CutMix, and EMA training loop and manually compared CutMix and Mixup variants, RandAugment, Mish activation, anti-aliased pooling, and wider networks across multiple random seeds. Selected the final 10-convolution CNN (87.18% validation accuracy, seed 43) and verified the packaged submission through load_model and predict on CPU and GPU.	35%
Lance Lam (2385064)	Developed and refined the model submission interface including model loading, image preprocessing, and batch prediction functions. Implemented testing procedures to verify prediction outputs, model	15%

	compatibility, and compliance with assignment requirements. Simplified and organized the submission code to improve readability and maintainability. Assisted with integrating the trained CNN into the final submission pipeline.	

How our Group Collaborated:

We divided responsibilities and shared updates/questions through group text messages. We used GitHub branches and commits to share code and coordinate changes together. We discussed validation results and worked together to decide which framework and model to use. We also coordinated the submission and contribution report through the same methods.